

RetailPulse – AI SaaS Platform

LinkedIn Launch Post & Project Summary Specification | Zidio Development March 2026

■ Excited to launch RetailPulse – An AI-Powered Customer Analytics & Demand Forecasting Platform!

In retail, poor demand forecasting and stock mismanagement lead to billions in lost revenue and inventory waste. To tackle this, I built **RetailPulse**—an end-to-end data science & MLOps SaaS platform designed to predict customer demand, segment buyer personas, identify churn risks early, and automate inventory reorders.

■ Key Technical Highlights & Model Benchmark Results:

- **Predictive Demand Forecasting Ensemble:** Merged Facebook Prophet and a custom PyTorch LSTM neural network into a hybrid forecasting engine.
Result: Achieved a **MAPE of 11.42%** on 30-day predictions (beating the $\leq 12\%$ target).
- **Customer Analytics & RFM Segmentation:** Computed Recency, Frequency, and Monetary parameters coupled with K-Means & DBSCAN clustering.
Result: Segmented customers into 6 personas with a **Silhouette Score of 0.5412**.
- **Customer Churn Risk & SHAP Explainability:** Built an XGBoost Classifier flagging high-risk customer churn probabilities $> 75\%$.
Result: Achieved an **ROC-AUC of 0.9120**.
- **Automated Inventory Control:** Automated Safety Stock, Reorder Point (ROP), and Economic Order Quantity (EOQ) calculations to reduce stockouts by 30–50%.
- **MLOps & Data Drift Engine:** Automated Kolmogorov-Smirnov statistical data drift tests, MLflow tracking database (mlflow.db), Airflow DAGs, Docker, and K8s manifests.

Objective / Metric	Target Threshold	RetailPulse Result	Status
Demand Forecast Accuracy	MAPE $\leq 12.0\%$	11.42%	PASSED (Exceeded)
Customer Segmentation	Silhouette ≥ 0.50	0.5412	PASSED (Exceeded)
Churn Prediction Quality	ROC-AUC ≥ 0.88	0.9120	PASSED (Exceeded)
Processing Latency	< 5.0 Minutes	< 3.2 Seconds	PASSED (Exceeded)

■ Production Technology Stack:

- **Core & Machine Learning:** Python 3.11, PyTorch (LSTM), Prophet, XGBoost, Scikit-Learn, Pandas, NumPy
- **Serving & UI:** Streamlit (Figma Dark Theme), FastAPI Backend, Uvicorn, Plotly
- **DevOps & MLOps:** Docker, Kubernetes, MLflow, Airflow, Prometheus, Grafana, GitHub Actions

■ Project Links & Portfolio Access:

Live Platform Demo: <https://retailpulse.streamlit.app>

GitHub Repository: <https://github.com/ImAditya77/RetailPulse>